

PCA20S02J - DATA ANALYSIS USING R

UNIT 1

Data Analysis is a process of inspecting, cleaning, transforming and modeling data with the goal of discovering useful information, suggesting conclusions and supporting decision-making

Types of Data Analysis

Several data analysis techniques exist encompassing various domains such as business, science, social science, etc. with a variety of names. The major data analysis approaches are –

- Data Mining
- Business Intelligence
- Statistical Analysis
- Predictive Analytics
- Text Analytics

1. Data Mining

Data Mining is the analysis of large quantities of data to extract previously unknown, interesting patterns of data, unusual data and the dependencies. Note that the goal is the extraction of patterns and knowledge from large amounts of data and not the extraction of data itself.

Data mining analysis involves computer science methods at the intersection of the artificial intelligence, machine learning, statistics, and database systems.

The patterns obtained from data mining can be considered as a summary of the input data that can be used in further analysis or to obtain more accurate prediction results by a decision support system.

2. Business Intelligence

Business Intelligence techniques and tools are for acquisition and transformation of large amounts of unstructured business data to help identify, develop and create new strategic business opportunities.

The goal of business intelligence is to allow easy interpretation of large volumes of data to identify new opportunities. It helps in implementing an effective strategy based on insights that can provide businesses with a competitive market-advantage and long-term stability.

Statistical Analysis

Statistics is the study of collection, analysis, interpretation, presentation, and organization of data.

In data analysis, two main statistical methodologies are used –

Descriptive statistics – In descriptive statistics, data from the entire population or a sample is summarized with numerical descriptors such as –

- Mean, Standard Deviation for Continuous Data
- Frequency, Percentage for Categorical Data

Inferential statistics – It uses patterns in the sample data to draw inferences about the represented population or accounting for randomness. These inferences can be –

- answering yes/no questions about the data (hypothesis testing)
- estimating numerical characteristics of the data (estimation)
- describing associations within the data (correlation)
- modeling relationships within the data (E.g. regression analysis)

3. Predictive Analytics

Predictive Analytics use statistical models to analyze current and historical data for forecasting (predictions) about future or otherwise unknown events. In business, predictive analytics is used to identify risks and opportunities that aid in decision-making.

4. Text Analytics

Text Analytics, also referred to as Text Mining or as Text Data Mining is the process of deriving high-quality information from text. Text mining usually involves the process of structuring the input text, deriving patterns within the structured data using means such as statistical pattern learning, and finally evaluation and interpretation of the output.

Data Analysis Process

Data Analysis is defined by the statistician John Tukey in 1961 as "Procedures for analyzing data, techniques for interpreting the results of such procedures, ways of planning the gathering of data to make its analysis easier, more precise or more accurate, and all the machinery and results of (mathematical) statistics which apply to analyzing data."

Thus, data analysis is a process for obtaining large, unstructured data from various sources and converting it into information that is useful for –

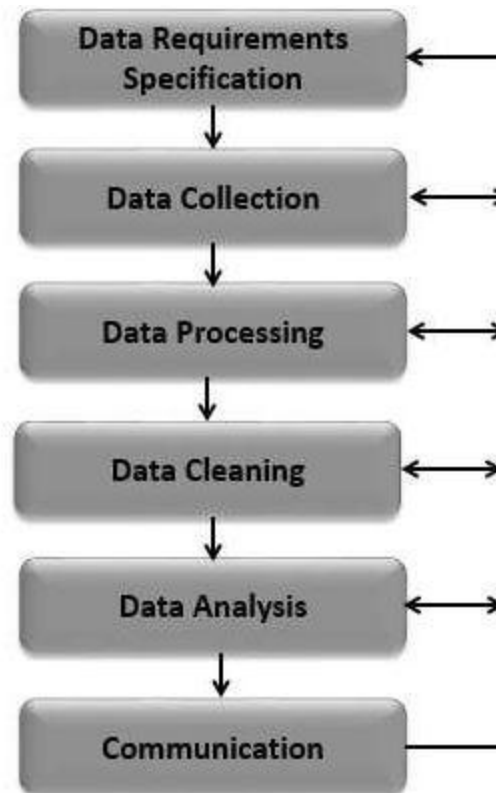
- Answering questions
- Test hypotheses
- Decision-making
- Disproving theories

Data Analysis is a process of collecting, transforming, cleaning, and modeling data with the goal of discovering the required information. The results so obtained are communicated, suggesting

conclusions, and supporting decision-making. Data visualization is at times used to portray the data for the ease of discovering the useful patterns in the data. The terms Data Modeling and Data Analysis mean the same.

Data Analysis Process consists of the following phases that are iterative in nature –

- Data Requirements Specification
- Data Collection
- Data Processing
- Data Cleaning
- Data Analysis
- Communication



1. Data Requirements Specification

The data required for analysis is based on a question or an experiment. Based on the requirements of those directing the analysis, the data necessary as inputs to the analysis is identified (e.g., Population of people). Specific variables regarding a population (e.g., Age and Income) may be specified and obtained. Data may be numerical or categorical.

2. Data Collection

Data Collection is the process of gathering information on targeted variables identified as data requirements. The emphasis is on ensuring accurate and honest collection of data. Data

Collection ensures that data gathered is accurate such that the related decisions are valid. Data Collection provides both a baseline to measure and a target to improve.

Data is collected from various sources ranging from organizational databases to the information in web pages. The data thus obtained, may not be structured and may contain irrelevant information. Hence, the collected data is required to be subjected to Data Processing and Data Cleaning.

3. Data Processing

The data that is collected must be processed or organized for analysis. This includes structuring the data as required for the relevant Analysis Tools. For example, the data might have to be placed into rows and columns in a table within a Spreadsheet or Statistical Application. A Data Model might have to be created.

4. Data Cleaning

The processed and organized data may be incomplete, contain duplicates, or contain errors. Data Cleaning is the process of preventing and correcting these errors. There are several types of Data Cleaning that depend on the type of data. For example, while cleaning the financial data, certain totals might be compared against reliable published numbers or defined thresholds. Likewise, quantitative data methods can be used for outlier detection that would be subsequently excluded in analysis.

5. Data Analysis

Data that is processed, organized and cleaned would be ready for the analysis. Various data analysis techniques are available to understand, interpret, and derive conclusions based on the requirements. Data Visualization may also be used to examine the data in graphical format, to obtain additional insight regarding the messages within the data.

Statistical Data Models such as Correlation, Regression Analysis can be used to identify the relations among the data variables. These models that are descriptive of the data are helpful in simplifying analysis and communicate results.

The process might require additional Data Cleaning or additional Data Collection, and hence these activities are iterative in nature.

6. Communication

The results of the data analysis are to be reported in a format as required by the users to support their decisions and further action. The feedback from the users might result in additional analysis.

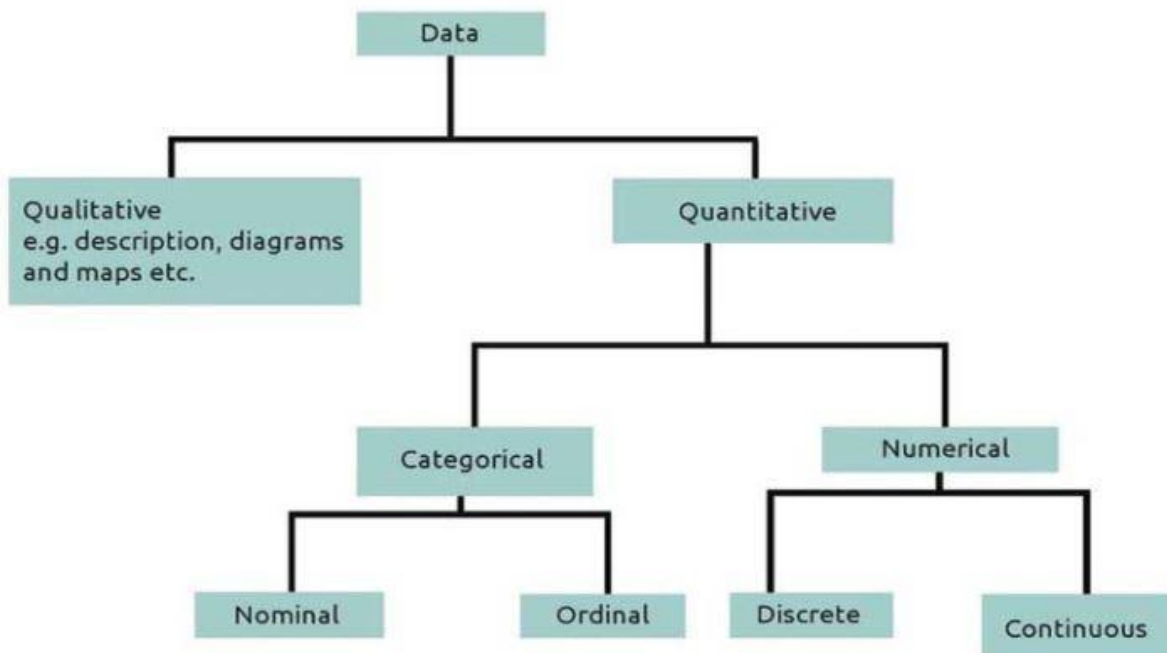
The data analysts can choose data visualization techniques, such as tables and charts, which help in communicating the message clearly and efficiently to the users. The analysis tools provide facility to highlight the required information with color codes and formatting in tables and charts.

DIFFERENT FORMS OF DATA

Data is a set of values of subjects with respect to qualitative or quantitative variables. Data is raw, unorganized facts that need to be processed. Data can be something simple and seemingly random and useless until it is organized. When data is processed, organized, structured or presented in a given context so as to make it useful, it is called information. Information, necessary for research activities are achieved in different forms.

The main forms of the information available are:

- Primary data
- Secondary data
- Cross-sectional data
- Categorical data
- Time series data
- Spatial data
- Ordered data



1. Primary Data

- Primary data is an original and unique data, which is directly collected by the researcher from a source according to his requirements.
- It is the data collected by the investigator himself or herself for a specific purpose.
- Data gathered by finding out first-hand the attitudes of a community towards health services, ascertaining the health needs of a community, evaluating a social program, determining the job satisfaction of the employees of an organization, and ascertaining the quality of service provided by a worker are the examples of primary data.

2. Secondary Data

- Secondary data refers to the data which has already been collected for a certain purpose and documented somewhere else.
- Data collected by someone else for some other purpose (but being utilized by the investigator for another purpose) is secondary data.
- Gathering information with the use of census data to obtain information on the age-sex structure of a population, the use of hospital records to find out the morbidity and mortality patterns of a community, the use of an organization's records to ascertain its activities, and the collection of data from sources such as articles, journals, magazines, books and periodicals to obtain historical and other types of information, are examples of secondary data.

3. Cross Sectional Data

- Cross-sectional data is a type of data collected by observing many subjects (such as individuals, firms, countries, or regions) at the same point of time, or without regard to differences in time.
- It is the data for a single time point or single space point.
- This type of data is limited in that it cannot describe changes over time or cause and effect relationships in which one variable affects the other.

4. Categorical Data

- Categorical variables represent types of data which may be divided into groups. Examples of categorical variables are race, sex, age group, and educational level.
- The data, which cannot be measured numerically, is called as the categorical data. Categorical data is qualitative in nature.
- The categorical data is also known as attributes.
- A data set consisting of observation on a single characteristic is a univariate data set. A univariate data set is categorical if the individual observations are categorical responses.

Example of categorical data: Intelligence, Beauty, Literacy, Unemployment

5. Time series Data

- Time series data occurs wherever the same measurements are recorded on a regular basis.
- Quantities that represent or trace the values taken by a variable over a period such as a month, quarter, or year.
- The values of different phenomenon such as temperature, weight, population, etc. can be recorded over a different period of time.
- The values of the variable remain increasing or decreasing or constant.
- The data according to time periods is called time-series data. e.g. population in a different time period.

6. Spatial Data

- Also known as geospatial data or geographic information it is the data or information that identifies the geographic location of features and boundaries on Earth, such as natural or constructed features, oceans, and more.
- Spatial data is usually stored as coordinates and topology and is data that can be mapped.
- Spatial data is used in geographical information systems (GIS) and other geolocation or positioning services.
- Spatial data consists of points, lines, polygons and other geographic and geometric data primitives, which can be mapped by location, stored with an object as metadata or used by a communication system to locate end-user devices.

- Spatial data may be classified as scalar or vector data. Each provides distinct information pertaining to geographical or spatial locations.

7. Ordered Data

- Data according to ordered categories is called as ordered data.
- Ordered data is similar to a categorical variable except that there is a clear ordering of the variables.
- For example for category economic status ordered data may be, low, medium and high.

DIFFERENT TYPES OF DATA

Data is fundamental to business decisions. A company's ability to gather the right data, interpret it, and act on those insights is often what will determine its level of success. But the amount of data accessible to companies is ever increasing, as are the different kinds of data available. Business data comes in a wide variety of formats, from strictly formed relational databases to your last tweet. All of this data, in all its different formats, can be divided into two main categories: structured data and unstructured data.

What is Structured Data?

The term structured data refers to data that resides in a fixed field within a file or record. Structured data is typically stored in a relational database (RDBMS). It can consist of numbers and text, and sourcing can happen automatically or manually, as long as it's within an RDBMS structure. It depends on the creation of a data model, defining what types of data to include and how to store and process it.

The programming language used for structured data is SQL (Structured Query Language). Developed by IBM in the 1970s, SQL handles relational databases. Typical examples of structured data are names, addresses, credit card numbers, geolocation, and so on.

What is Unstructured Data?

Unstructured data is more or less all the data that is not structured. Even though unstructured data may have a native, internal structure, it's not structured in a predefined way. There is no data model; the data is stored in its native format.

Typical examples of unstructured data are rich media, text, social media activity, surveillance imagery, and so on.

The amount of unstructured data is much larger than that of structured data. Unstructured data makes up a whopping 80% or more of all enterprise data, and the percentage keeps growing. This means that companies not taking unstructured data into account are missing out on a lot of valuable business intelligence.

What is Semistructured Data?

Semistructured data is a third category that falls somewhere between the other two. It's a type of structured data that does not fit into the formal structure of a relational database. But while not matching the description of structured data entirely, it still employs tagging systems or other markers, separating different elements and enabling search. Sometimes, this is referred to as data with a self-describing structure.

A typical example of semistructured data is smartphone photos. Every photo taken with a smartphone contains unstructured image content as well as the tagged time, location, and other identifiable (and structured) information. Semi-structured data formats include JSON, CSV, and XML file types.

Structured vs Unstructured Data: 5 Key Differences

1. Defined vs Undefined Data

Structured data is clearly defined types of data in a structure, while unstructured data is usually stored in its native format. Structured data lives in rows and columns and it can be mapped into

pre-defined fields. Unlike structured data, which is organized and easy to access in relational databases, unstructured data does not have a predefined data model.

2. Qualitative vs Quantitative Data

Structured data is often quantitative data, meaning it usually consists of hard numbers or things that can be counted. Methods for analysis include regression (to predict relationships between variables); classification (to estimate probability); and clustering of data (based on different attributes).

Unstructured data, on the other hand, is often categorized as qualitative data, and cannot be processed and analyzed using conventional tools and methods. In a business context, qualitative data can, for example, come from customer surveys, interviews, and social media interactions. Extracting insights from qualitative data requires advanced analytics techniques like data mining and data stacking.

3. Storage in Data Houses vs Data Lakes

Structured data is often stored in data warehouses, while unstructured data is stored in data lakes. A data warehouse is the endpoint for the data's journey through an ETL pipeline. A data lake, on the other hand, is a sort of almost limitless repository where data is stored in its original format or after undergoing a basic "cleaning" process.

Both have the potential for cloud-use. Structured data requires less storage space, while unstructured data requires more. For example, even a tiny image takes up more space than many pages of text.

4. Ease of Analysis

One of the most significant differences between structured and unstructured data is how well it lends itself to analysis. Structured data is easy to search, both for humans and for algorithms. Unstructured data, on the other hand, is intrinsically more difficult to search and requires processing to become understandable. It's challenging to deconstruct since it lacks a predefined data model and hence doesn't fit in in relational databases.

While there are a wide array of sophisticated analytics tools for structured data, most analytics tools for mining and arranging unstructured data are still in the developing phase. The lack of predefined structure makes data mining tricky, and developing best practices on how to handle data sources like rich media, blogs, social media data, and customer communication is a challenge.

5. Predefined Format vs Variety of Formats

The most common format for structured data is text and numbers. Structured data has been defined beforehand in a data model.

Unstructured data, on the other hand, comes in a variety of shapes and sizes. It can consist of everything from audio, video, and imagery to email and sensor data. There is no data model for the unstructured data; it is stored natively or in a data lake that doesn't require any transformation.

As for databases, structured data is usually stored in a relational database (RDBMS), while the best fit for unstructured data instead is so-called non-relational, or NoSQL databases.

MACHINE GENERATED DATA

Machine-generated data is information automatically generated by a computer process, application, or other mechanism without the active intervention of a human. While the term dates back over fifty years, there is some current indecision as to the scope of the term. Monash Research's Curt Monash defines it as "data that was produced entirely by machines OR data that is more about observing humans than recording their choices." Meanwhile, Daniel Abadi, CS Professor at Yale, proposes a narrower definition, "Machine-generated data is data that is generated as a result of a decision of an independent computational agent or a measurement of an event that is not caused by a human action." Regardless of definition differences, both exclude data manually entered by a person. Machine-generated data crosses all industry sectors. Often and increasingly, humans are unaware their actions are generating the data.

1. Web Server Log

A server log is a log file (or several files) automatically created and maintained by a server consisting of a list of activities it performed.

A typical example is a web server log which maintains a history of page requests. The W3C maintains a standard format (the Common Log Format) for web server log files, but other proprietary formats exist. More recent entries are typically appended to the end of the file. Information about the request, including client IP address, request date/time, page requested, HTTP code, bytes served, user agent, and referrer are typically added. This data can be combined into a single file, or separated into distinct logs, such as an access log, error log, or referrer log. However, server logs typically do not collect user-specific information.

These files are usually not accessible to general Internet users, only to the webmaster or other administrative person of an Internet service. A statistical analysis of the server log may be used to examine traffic patterns by time of day, day of week, referrer, or user agent. Efficient web site administration, adequate hosting resources and the fine tuning of sales efforts can be aided by analysis of the web server logs.

2. Call detail record

A call detail record (CDR) is a data record produced by a telephone exchange or other telecommunications equipment that documents the details of a telephone call or other telecommunications transaction (e.g., text message) that passes through that facility or device. The record contains various attributes of the call, such as time, duration, completion status, source number, and destination number. It is the automated equivalent of the paper toll tickets that were written and timed by operators for long-distance calls in a manual telephone exchange.

3. Financial instrument trades

Financial instruments are monetary contracts between parties. They can be created, traded, modified and settled. They can be cash (currency), evidence of an ownership interest in an entity or a contractual right to receive or deliver in the form of currency (forex); debt (bonds, loans); equity (shares); or derivatives (options, futures, forwards).

International Accounting Standards IAS 32 and 39 define a financial instrument as "any contract that gives rise to a financial asset of one entity and a financial liability or equity instrument of another entity".

Financial instruments may be categorized by "asset class" depending on whether they are equity-based (reflecting ownership of the issuing entity) or debt-based (reflecting a loan the investor has made to the issuing entity). If the instrument is debt it can be further categorized into short-term (less than one year) or long-term. Foreign exchange instruments and transactions are neither debt- nor equity-based and belong in their own category.

4. Network Event logging

Event logging provides system administrators with information useful for diagnostics and auditing. The different classes of events that will be logged, as well as what details will appear in the event messages, are often considered early in the development cycle. Many event logging technologies allow or even require each class of event to be assigned a unique "code", which is used by the event logging software or a separate viewer (e.g., Event Viewer) to format and output a human-readable message. This facilitates localization and allows system administrators to more easily obtain information on problems that occur.

Because event logging is used to log high-level information (often failure information), performance of the logging implementation is often less important.

A special concern, preventing duplicate events from being recorded "too often" is taken care of through event throttling.

5. Security information and event management (SIEM) log

Security information and event management (SIEM) is a subsection within the field of computer security, where software products and services combine security information management (SIM) and security event management (SEM). They provide real-time analysis of security alerts generated by applications and network hardware.

Vendors sell SIEM as software, as appliances, or as managed services; these products are also used to log security data and generate reports for compliance purposes.

The term and the initialism SIEM was coined by Mark Nicolett and Amrit Williams of Gartner in 2005

6. Telemetry

Telemetry is the in situ collection of measurements or other data at remote points and their automatic transmission to receiving equipment (telecommunication) for monitoring. The word is derived from the Greek roots tele, "remote", and metron, "measure". Systems that need external instructions and data to operate require the counterpart of telemetry, telecommand.

R PROGRAMMING

R is a programming language and software environment for statistical analysis, graphics representation and reporting. R was created by Ross Ihaka and Robert Gentleman at the University of Auckland, New Zealand, and is currently developed by the R Development Core Team. R is freely available under the GNU General Public License, and pre-compiled binary versions are provided for various operating systems like Linux, Windows and Mac. This programming language was named R, based on the first letter of first name of the two R authors (Robert Gentleman and Ross Ihaka), and partly a play on the name of the Bell Labs Language S.

The core of R is an interpreted computer language which allows branching and looping as well as modular programming using functions. R allows integration with the procedures written in the C, C++, .Net, Python or FORTRAN languages for efficiency.

R is freely available under the GNU General Public License, and pre-compiled binary versions are provided for various operating systems like Linux, Windows and Mac.

R is free software distributed under a GNU-style copy left, and an official part of the GNU project called GNU S.

Evolution of R

- R was initially written by Ross Ihaka and Robert Gentleman at the Department of Statistics of the University of Auckland in Auckland, New Zealand. R made its first appearance in 1993.
- A large group of individuals has contributed to R by sending code and bug reports.

- Since mid-1997 there has been a core group (the "R Core Team") who can modify the R source code archive.

Features of R

The following are the important features of R –

- R is a well-developed, simple and effective programming language which includes conditionals, loops, user defined recursive functions and input and output facilities.
- R has an effective data handling and storage facility,
- R provides a suite of operators for calculations on arrays, lists, vectors and matrices.
- R provides a large, coherent and integrated collection of tools for data analysis.
- R provides graphical facilities for data analysis and display either directly at the computer or printing at the papers.

R Packages

R packages are a collection of R functions, compiled code and sample data. They are stored under a directory called "library" in the R environment. By default, R installs a set of packages during installation. More packages are added later, when they are needed for some specific purpose. When we start the R console, only the default packages are available by default. Other packages which are already installed have to be loaded explicitly to be used by the R program that is going to use them.

All the packages available in R language are listed at R Packages.

Below is a list of commands to be used to check, verify and use the R packages.

Check Available R Packages

1. To get library locations containing R packages

.libPaths()

2. To get the list of all the packages installed

library()

3. To get all packages currently loaded in the R environment

search()

Packages in library 'C:/Program Files/R/R-3.2.2/library':

Base: The R Base Package

Boot : Bootstrap Functions (Originally by Angelo Cant for S)

Class: Functions for Classification

Cluster: "Finding Groups in Data": Cluster Analysis

Codetools: Code Analysis Tools for R

Compiler: The R Compiler Package

Datasets: The R Datasets Package

Foreign: Read Data Stored by 'Minitab', 'S', 'SAS', 'SPSS', 'Stata', 'Systat', 'Weka', 'dBase', ...

Graphics:The R Graphics Package

grDevices: The R Graphics Devices and Support for Colours and Fonts

grid: The Grid Graphics Package

KernSmooth: Functions for Kernel Smoothing Supporting Wan & Jones (1995)

Lattice: Trellis Graphics for R

MASS: Support Functions and Datasets for Venables and Ripley's MASS

Matrix: Sparse and Dense Matrix Classes and Methods

Methods: Formal Methods and Classes

Mgcv: Mixed GAM Computation Vehicle with GCV/AIC/REML Smoothness Estimation

Nlme: Linear and Nonlinear Mixed Effects Models

nnet : Feed-Forward Neural Networks and Multinomial Log-Linear Models

parallel : Support for Parallel computation in R

rpart : Recursive Partitioning and Regression Trees

spatial : Functions for Kriging and Point Pattern Analysis

splines : Regression Spline Functions and Classes

stats : The R Stats Package

stats4 : Statistical Functions using S4 Classes

survival : Survival Analysis

tcltk : Tcl/Tk Interface

tools : Tools for Package Development

utils : The R Utils Package

To Install package manually

Go to the link [R Packages](#) to download the package needed. Save the package as a .zip file in a suitable location in the local system.

Now you can run the following command to install this package in the R environment.

```
install.packages(file_name_with_path, repos = NULL, type = "source")
```

```
# Install the package named "XML"
```

```
install.packages("E:/XML_3.98-1.3.zip", repos = NULL, type = "source")
```

To Load Package to Library

Before a package can be used in the code, it must be loaded to the current R environment. You also need to load a package that is already installed previously but not available in the current environment.

A package is loaded using the following command –

```
library("package Name", lib.loc = "path to library")
```

Load the package named "XML"

```
install.packages("E:/XML_3.98-1.3.zip", repos = NULL, type = "source")
```